Rajeev Jain

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Principal Research Software Engineer

Summary

Research software engineer at Argonne National Laboratory since 2009. I build scientific software that runs on DOE supercomputers, and lately the tooling that lets AI agents use it. Lead developer of UXarray and its MCP server; contributor to E3SM's SimBoard. Two R&D 100 Awards.

Experience

Argonne National Laboratory **2009 – Present**
Principal Specialist, Research Software Engineering *Mathematics and Computer Science Division, Lemont, IL*

- **SimBoard (E3SM).** Contribute to the web catalog of E3SM simulation runs: case metadata, provenance, and links to diagnostics. FastAPI, PostgreSQL, React, deployed at NERSC.
- **UXarray.** Lead developer of a Python library for unstructured climate grids, used at DOE labs and NCAR. Made a 300-million-cell, 63 GB mesh loadable in 306 s; fixed 0.68 m of geometric drift with error-free arithmetic; 1.67× faster in a Numba hot path.
- **UXarray MCP server.** Wrote an MCP server that lets AI agents run UXarray analyses on HPC systems through Globus Compute. Presented at SciFM26 and IEEE eScience 2026.
- **SPOTTER-AI (DOE Genesis Mission).** Attention capture and poisoning experiments on cancer drug-response models, for provenance-based integrity checks.
- **SFNO on Aurora.** Ported a PyTorch climate emulator to Intel GPUs and set up distributed training.
- **CANDLE and IMPROVE.** Built a hyperparameter search workflow that ran 10,000+ training jobs on Summit, Theta, and Cori; led the IMPROVE benchmark framework. R&D 100 Award, 2023.
- **FLASH-X.** Asynchronous HDF5 checkpointing with compression, cutting checkpoint overhead 40–70% on Summit. R&D 100 Award, 2022.
- **MeshKit (PI, DOE NEAMS).** Reactor core mesh generation; a 101-million-element core mesh in about 7 minutes on 712 cores.
- **Maintenance and review.** Maintain the ncvis NetCDF viewer (C++); added the Argo model provider to the E3SM AI platform; regular reviewer on UXarray.

University of Chicago **2023 – Present**
Staff At-Large *Consortium for Advanced Science and Engineering (CASE)*

- Joint appointment for cancer and Earth science software collaborations.

Arizona State University **2007 – 2009**
Research and Teaching Assistant *Structural and Computational Mechanics Lab*

- Finite element research on blast mitigation.

Skills

Languages: Python, C++, Fortran, Bash
Scientific Python: NumPy, Xarray, Numba, Dask, Polars, PyTorch
AI systems: MCP servers, agent evaluation, distributed training, hyperparameter search
HPC: MPI, HDF5, NetCDF, SZ3 and ZFP compression, Intel and NVIDIA GPUs, PBS, Slurm
Engineering: CMake, CI/CD, packaging (conda-forge, PyPI), Globus Compute, code review

Awards

- R&D 100 Award, CANDLE (2023). R&D 100 Award, FLASH-X (2022).
- Best Paper, International Meshing Roundtable (2010). ATPESC scholar (2015).
- Program committee, AgenticAI4HPC at SC26 and AGENT4SC at IEEE eScience 2026. DOE SBIR/STTR proposal reviewer.

Selected Publications

- **Jain, R.** and Jacob, R. "Beyond Tool Execution: Evaluating Scientific MCP Interfaces with UXarray." *AGENT4SC at IEEE eScience*, 2026.
- Partin, A., …, **Jain, R.**, et al. "Benchmarking community drug response prediction models." *Briefings in Bioinformatics*, 2026.
- **Jain, R.**, Tang, H., Dhruv, A., and Byna, S. "Enabling Data Reduction for Flash-X Simulations." *DRBSD-10 at SC24*, 2024.
- **Jain, R.**, Wozniak, J. M., et al. "Cross-HPO: Optimizing Neural Networks for Cancer Drug Response." *CAFCW at SC24*, 2024.
- Wozniak, J. M., **Jain, R.**, et al. "CANDLE/Supervisor: A workflow framework for machine learning applied to cancer research." *BMC Bioinformatics*, 2018.

Full list: Google Scholar.

Education

University of Chicago 2020
M.S. in Computer Science

Arizona State University 2009
M.S. in Structural Engineering

Indian Institute of Technology (ISM) Dhanbad 2006
B.Tech. in Mechanical Engineering